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Display Device Having an Integrated Wide Angle Camera System

Abstract

A display device which includes a display component; a display frame; an embedded camera system, the embedded camera system including a camera component, the camera component being physically coupled to the display frame to embed the camera component within the display device and a lens component, the lens component comprising a normal field of view lens portion and a wide angle field of view lens portion.

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Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

[0001] The present invention relates to information handling systems. More specifically,

embodiments of the invention relate to information handling system display devices.

Description of the Related Art

[0002] As the value and use of information continues to increase, individuals and businesses seek additional ways to process and store information. One option available to users is information handling systems. An information handling system generally processes, compiles, stores, and/or communicates information or data for business, personal, or other purposes thereby allowing users to take advantage of the value of the information. Because technology and information handling needs and requirements vary between different users or applications, information handling systems may also vary regarding what information is handled, how the information is handled, how much information is processed, stored, or communicated, and how quickly and efficiently the information may be processed, stored, or communicated. The variations in information handling systems allow for information handling systems to be general or configured for a specific user or specific use such as financial transaction processing, airline reservations, enterprise data storage, or global communications. In addition, information handling systems may include a variety of hardware and software components that may be configured to process, store, and communicate information and may include one or more computer systems, data storage systems, and networking systems.

SUMMARY OF THE INVENTION

[0003] An information handling system display device which includes an integrated camera system is disclosed.

[0004] More specifically, in one embodiment the invention relates to an embedded camera system for a display device comprising: a camera component, the camera component being physically coupled to a display frame to embed the camera component within the display device; and, a lens component, the lens component comprising a normal field of view lens portion and a wide angle field of view lens portion.

[0005] In another embodiment the invention relates to a display device comprising: a display component; a display frame; an embedded camera system, the embedded camera system comprising a camera component, the camera component being physically coupled to a display frame to embed the camera component within the display device; and, a lens component, the lens component comprising a normal field of view lens portion and a wide angle field of view lens portion.

[0006] In another embodiment the invention relates to a system comprising: a processor; a data bus coupled to the processor; and a display device, the display device comprising an embedded camera system, the embedded camera system comprising a camera component, the camera component being physically coupled to a display frame to embed the camera component within the display device; and, a lens component, the lens component comprising a normal field of view lens portion and a wide angle field of view lens portion.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] The present invention may be better understood, and its numerous objects, features and advantages made apparent to those skilled in the art by referencing the accompanying drawings. The use of the same reference number throughout the several figures designates a like or similar element.

[0008] FIG. 1 shows a general illustration of components of an information handling system as implemented in the present disclosure.

[0009] FIG. 2 shows a block diagram of an information handling system configuration and fabrication environment.

[0010] FIGS. 3A and 3B, generally referred to as FIG. 3, respectively show a front view and an

exploded perspective view of a display device having an embedded camera system.

[0011] FIG. 4 shows a cut away side view of a portion of a display device having an embedded camera system.

[0012] FIGS. 5A and 5B, generally referred to as FIG. 5, respectively show front and side views of a wide angle camera system.

[0013] FIGS. 6A and 6B, generally referred to as FIG. 6, respectively show examples of a wide angle shaded image and a wide angle corrected image.

[0014] FIGS. 7A and 7B, generally referred to as FIG. 7, respectively show examples of a wide angle distorted image and a wide angle corrected image.

DETAILED DESCRIPTION

[0015] Certain aspects of the present disclosure include an appreciation that camera systems often have an associated field of view. As used herein, field of view refers to an open observable area of an optical device such as a camera system. Accordingly, a field of view represents how much of an area a device can observe. Various aspects of the present disclosure include an appreciation that a normal field of view often ranges from a 65 degree field of view to a 90 degree field of view. Various aspects of the present disclosure include an appreciation that a wide angle field of view often includes a field of view of substantially (i.e., $\pm 20\%$) 120 degrees. Various aspects of the present disclosure include an appreciation that a 65 degree field of view provides an optical device with a view of a user positioned in front of the optical device, a 78 degree field of view provides an optical device with a view of a user positioned in front of the optical device as well as a desk space surrounding the user and a 90 degree field of view provides an optical device with a view of a user positioned in front of the optical device as well as additional space surrounding the user. Various aspects of the present disclosure include an appreciation that a 90 degree field of view is often referred to as a wide angle field of view. Various aspects of the present disclosure include an appreciation that a wide angle field of view is often used to provide a conferencing function. Various aspects of the present disclosure include an appreciation that a conferencing function provides an optical device with a field of view which allows a simultaneous view of a plurality of user positioned in front of the optical device.

[0016] Certain aspects of the disclosure reflect an appreciation that it would be desirable to provide display devices which include embedded camera systems with a wide angle field of view function (also referred to as a wide angle function). Certain aspects of the disclosure reflect an appreciation that certain known embedded camera systems include embedded lens systems. Certain aspects of the disclosure reflect an appreciation that it is challenging to provide embedded lens systems to certain known display devices, such as narrow bezel display devices. With certain narrow bezel display devices there is not enough space on the bezel to allow for a wide angle field of view lens system.

[0017] Accordingly, a system and method are disclosed for providing an information handling system display device with an embedded camera system having an embedded wide angle lens system. In various embodiments, the embedded camera system includes a camera component, lens component, a display frame portion, a display device frame portion, or a combination thereof. In certain embodiments, the camera component is embedded along an edge of the display device frame portion. In certain embodiments, the camera component is embedded along a top edge of the display device frame portion. In certain embodiments, the camera component is embedded in a center area of the display device frame portion.

[0018] In certain embodiments, the lens system is mounted along an edge of the display device frame portion. In certain embodiments, the lens system is mounted along a top edge of the display device frame portion. In certain embodiments, the lens system is mounted in a center area of the display device frame portion. In certain embodiments, the lens system is mounted in front of the camera component.

[0019] In certain embodiments, the lens system includes a normal lens portion and a wide angle

field of view portion. In certain embodiments, the wide angle field of view portion includes a metasurface component. In certain embodiments, the camera system includes a camera control system. In certain embodiments, the camera control system includes a compensation management module. In certain embodiments, the compensation management module manages distortion produced by the wide angle field of view portion, shading irregularities produced by the wide angle field of view portion, or a combination thereof. In certain embodiments, the compensation management module includes an image signal processor.

[0020] FIG. 1 is a generalized illustration of an information handling system **100** that can be used to implement the system and method of the present invention. The information handling system **100** includes a processor (e.g., central processor unit or “CPU”) **102**, input/output (I/O) devices **104**, such as a display, a keyboard, a mouse, a touchpad or touchscreen, and associated controllers, a hard drive or disk storage **106**, and various other subsystems **108**. In various embodiments, the information handling system **100** also includes network port **110** operable to connect to a network **140**, which is likewise accessible by a service provider server **142**. The information handling system **100** likewise includes system memory **112**, which is interconnected to the foregoing via one or more buses **114**. System memory **112** further comprises operating system (OS) **116**. In various embodiments, the information handling system **100** is coupled with and communicates with a display device **120**. In certain embodiments, system memory **112** includes a camera control system **150**, a camera compensation management system **152**, or a combination thereof. In various embodiments, the information handling system is integrated with the display device to provide an all in one (AIO) type information handling system. In certain embodiments, the information handling system **100** includes a portable information handling system which includes an integrated display device.

[0021] For purposes of this disclosure, an information handling system may include any instrumentality or aggregate of instrumentalities operable to compute, classify, process, transmit, receive, retrieve, originate, switch, store, display, manifest, detect, record, reproduce, handle, or utilize any form of information, intelligence, or data for business, scientific, control, or other purposes. For example, an information handling system may be a personal computer, a network storage device, or any other suitable device and may vary in size, shape, performance, functionality, and price. The information handling system may include random access memory (RAM), one or more processing resources such as a central processing unit (CPU) or hardware or software control logic, ROM, and/or other types of nonvolatile memory. Additional components of the information handling system may include one or more disk drives, one or more network ports for communicating with external devices as well as various input and output (I/O) devices, such as a keyboard, a mouse, and a video display. The information handling system may also include one or more buses operable to transmit communications between the various hardware components.

[0022] In certain embodiments, the computer system **100**, the display device **120**, or a combination thereof, include an embedded camera system. In various embodiments, the embedded camera system includes a camera component, lens component, a display frame portion, a display device frame portion, or a combination thereof. In certain embodiments, the camera component is embedded along an edge of the display device frame portion. In certain embodiments, the camera component is embedded along a top edge of the display device frame portion. In certain embodiments, the camera component is embedded in a center area of the display device frame portion.

[0023] In certain embodiments, the lens system is mounted along an edge of the display device frame portion. In certain embodiments, the lens system is mounted along a top edge of the display device frame portion. In certain embodiments, the lens system is mounted in a center area of the display device frame portion. In certain embodiments, the lens system is mounted in front of the camera component.

[0024] In certain embodiments, the lens system includes a normal lens portion and a wide angle

field of view lens portion. In certain embodiments, the wide angle field of view portion includes a metasurface component.

[0025] In certain embodiments, the camera system includes a camera control system. In certain embodiments, the camera control system **150** includes the compensation management system **152**. In certain embodiments, the camera control system **150** controls a receiving field of view of the camera component. In certain embodiments, the camera control system **150** electronically activates the wide angle field of view lens portion. In certain embodiments, the compensation management module manages image irregularities produced by the wide angle field of view lens portion. In certain embodiments, the compensation management module manages distortion produced by the wide angle field of view lens portion, shading irregularities produced by the wide angle field of view portion, or a combination thereof. In certain embodiments, the compensation management module includes an image signal processor.

[0026] FIG. **2** is a block diagram of an information handling system configuration and fabrication environment **200** implemented in accordance with an embodiment of the invention. In certain embodiments, the information handling system configuration and fabrication environment **200** may include a repository of information handling system configuration and fabrication data **220**. In certain embodiments, the repository of information handling system configuration and fabrication data **220** may be local or may be executed remotely.

[0027] In certain embodiments, the user device **204** is used to exchange information between the user **202** and a product configuration system **250**, and a custom product fabrication system **250**, through the use of a network **240**. In certain embodiments, the network **240** may be a public network, such as a public internet protocol (IP) network, a physical private network, a wireless network, a virtual private network (VPN), or any combination thereof. Skilled practitioners of the art will recognize that many such embodiments are possible and the foregoing is not intended to limit the spirit, scope or intent of the invention. As used herein, a user device **204** refers to an information handling system such as a personal computer, a laptop computer, a tablet computer, a personal digital assistant (PDA), a smart phone, a mobile telephone, or other device that is capable of communicating and processing data. In certain embodiments, the user device **204** may be configured to present a configuration and/or fabrication system user interface (UI) **240**. In certain embodiments, the configuration and/or fabrication system UI **240** may be implemented to present a graphical representation **242** of configuration and/or fabrication information.

[0028] In various embodiments, the configuration and/or fabrication system UI **240** may be presented via a website. In certain embodiments, the website may be provided by the product configuration system **250**. For the purposes of this disclosure a website may be defined as a collection of related web pages which are identified with a common domain name and is published on at least one web server. A website may be accessible via a public IP network or a private local network.

[0029] A web page is a document which is accessible via a browser which displays the web page via a display device of an information handling system. In various embodiments, the web page also includes the file which causes the document to be presented via the browser. In various embodiments, the web page may comprise a static web page, which is delivered exactly as stored and a dynamic web page, which is generated by a web application that is driven by software that enhances the web page via user input to a web server.

[0030] In certain embodiments, the user device **204** may be implemented to interact with the product configuration system **250**, which in turn may be executing on a separate information handling system **200**. In various embodiments, the product configuration system **250** interacts with a custom product fabrication system **252**. In various embodiments, the custom product fabrication system **252** fabricates products.

[0031] In various embodiments, the fabricated product includes a display device. In various embodiments, the display device includes an embedded camera system. In various embodiments,

the embedded camera system includes a camera component, a lens component, a display device frame portion, or a combination thereof.

[0032] In certain embodiments, the camera component is embedded along an edge of the display device frame portion. In certain embodiments, the camera component is embedded along a top edge of the display device frame portion. In certain embodiments, the camera component is embedded in a center area of the display device frame portion.

[0033] FIGS. 3A and 3B, generally referred to as FIG. 3, respectively show a front view and an exploded perspective view of a display device **300** having an embedded camera system **310**. In certain embodiments, the display device **300** includes a display frame component **312**, a frame component **314**, a front deco component **316**, or a combination thereof. In certain embodiments, a display **318** is mounted to the display frame component **312**. In certain embodiments, the display device frame component **314** is integrated within a middle frame component of the display device **300**. In certain embodiments, the front deco component **350** is mounted along an edge of the display frame component.

[0034] In certain embodiments, the embedded camera system **310** includes a camera component **320** having an associated lens component, a display device frame portion **330**, a display device frame portion **340**, or a combination thereof. In certain embodiments, the display device frame portion **330** is integrated within the display frame component **312** of the display device **300**. In certain embodiments, the display device frame portion **340** is integrated within the display device frame component **314** of the display device **300**.

[0035] In certain embodiments, the camera component **320** is embedded along an edge of the display device frame portion **340**. In certain embodiments, the camera component **320** fits within a notch defined by the display device frame portion **330**. In certain embodiments, the camera component **320** is embedded along a top edge of the display device frame portion **340**. In certain embodiments, the camera component **320** is embedded in a center area of the display device frame portion **340**.

[0036] In certain embodiments, the lens component is mounted along an edge of the display frame portion **330**. In certain embodiments, the associated electronic lens component is mounted along a top edge of the display frame portion **330**. In certain embodiments, the camera component **320**, the associated lens component or a combination thereof, are mounted in a center area of the display device frame portion.

[0037] FIG. 4 shows a cut away side view of a portion of a display device **400** having an embedded camera system **405**. In certain embodiments, display device **400** corresponds to display device **300**. In certain embodiments, the embedded camera system **405** corresponds to embedded camera system **310**.

[0038] In certain embodiments, the display device **400** includes a display portion **412**, a support frame portion **414**, a rear frame portion **416**, or a combination thereof. In certain embodiments, the display frame portion **412** includes a display component **420**, a front display panel **422**, or a combination thereof. In certain embodiments, one or both the display component **420** and the front display panel **422** are mounted to the display frame portion **412**. In certain embodiments, the display device frame portion **414** is integrated within a middle frame component of the display device **400**. In certain embodiments, the rear frame portion **416** is mounted along an outside edge of the support frame portion **414**.

[0039] In certain embodiments, the embedded camera system **405** includes a camera component **430**, an associated lens component **432**, or a combination thereof. In certain embodiments, the camera component **430** is embedded along an edge of the display device frame portion **440**. In certain embodiments, the camera component **430** fits within a void defined by the display frame. In certain embodiments, the camera component **430** is embedded along a top edge of the display device frame. In certain embodiments, the camera component **430** is embedded in a center area of the display device frame.

[0040] In certain embodiments, the lens component **432** is mounted in front of the camera component **430**. In certain embodiments, the lens component **432** is positioned along a top edge of the front panel display **422**. In certain embodiments, the camera component **430**, the lens component **432** or a combination thereof, are mounted in a center area of the display device frame. [0041] In certain embodiments, the lens component **432** includes a normal field of view lens portion and a wide angle field of view lens portion. As used herein, a normal field of view lens portion broadly refers to a lens component which provides a field of view from substantially (i.e., $\pm 20\%$) 65 degrees to substantially (i.e., $\pm 20\%$) 90 degrees. As used herein, a wide angle field of view lens portion broadly refers to a lens component which provides a field of view greater than substantially (i.e., $\pm 20\%$) 90 degrees. In certain embodiments, the wide angle field of view lens portion provides a camera system with a conferencing function.

[0042] In certain embodiments, the normal field of view lens portion is constructed from plastic, glass, or a combination thereof. In certain embodiments, the wide angle field of view portion includes a metasurface component. In certain embodiments, the embedded camera system **430** includes an associated camera control system. In certain embodiments, the associated camera control system includes a compensation management module. In certain embodiments, the compensation management module manages distortion produced by the wide angle field of view portion, shading irregularities produced by the wide angle field of view portion, or a combination thereof. In certain embodiments, the compensation management module includes an image signal processor.

[0043] FIGS. 5A and 5B, generally referred to as FIG. 5, respectively show front and side views of a wide angle field of view camera system **500**. In certain embodiments, the wide angle field of view camera system **500** includes a camera component **510**, a lens component **520**, or a combination thereof. In certain embodiments, the lens component **520** includes a normal field of view lens portion **530**, a wide angle field of view lens portion **532**, or a combination thereof. In certain embodiments, the lens component **520** further includes a filter glass portion **534**, a lens edge portion **536**, or a combination thereof. In certain embodiments, the lens edge portion **534** includes a black matrix which defines an outside edge of the lens component **520**. In certain embodiments, the normal field of view lens portion **530**, the wide angle field of view lens portion **532**, the lens edge portion **534**, or a combination thereof, are substantially (i.e., $\pm 20\%$) circularly shaped. In certain embodiments, the wide angle field of view lens portion **532** encircles the normal field of view lens portion **530**. In certain embodiments, an outside edge of the normal field of view lens portion **530** is contiguous with an inside edge of the wide angle field of view lens portion **532**.

[0044] In certain embodiments, the lens component **520** is configured with a first predetermined diameter. In certain embodiments, the normal field of view lens portion **530** is configured with a second predetermined diameter. In certain embodiments, the second predetermined diameter is smaller than the first predetermined diameter. In certain embodiments, the first predetermined diameter is a combination of the second predetermined diameter and an additional diameter provided by the wide angle field of view lens portion **532**. In certain embodiments, the additional diameter provided by the wide angle field of view lens portion **532** is substantially smaller (i.e., $\pm 15\%$) than the second predetermined diameter.

[0045] In certain embodiments, the wide angle field of view lens portion **532** is constructed to include a metasurface component. In certain embodiments, the metasurface component is mounted on color filter glass portion **534** of the lens component **520**. As used herein, a metasurface broadly refers to a sheet material having a sub-wavelength thickness which controls behaviors of electromagnetic waves. In certain embodiments, the metasurface includes plasmonics which provide an ultrathin layer of metamaterials which control a wavefront of optical beams. Controlling a wavefront of electromagnetic waves enables the metasurface to impart local, gradient phase shifts to the incoming waves such that the metasurface provides a lens function.

[0046] In certain embodiments, the metasurface component includes a plurality of unit cells. As

used herein, a unit cell of a metasurface broadly refers to a metasurface subcomponent which includes a plurality of silicon nanodisks. In certain embodiments, the plurality of nanodisks of the metasurface subcomponent are represented by a plurality of parameters. In certain embodiments, the plurality of parameters includes a diameter parameter and a height parameter of each nanodisk, or a combination thereof. In certain embodiments, the plurality of parameters includes a gap parameter representing a gap between two nanodisks within a unit cell, a period parameter representing a period of repeating unit cells, or a combination thereof. In certain embodiments, the plurality of parameters may be adjusted to achieve a desired surface refractive index. In certain embodiments, the plurality of parameters are adjusted to control the sizes and shapes of the plurality of unit cells are adjusted to achieve a desired surface refractive index. In certain embodiments, unit cell sizes and shapes of the wide field of view lens portion **532** are adjusted so that functions can be patterned on the surface of the metasurface.

[0047] FIGS. **6A** and **6B**, generally referred to as FIG. **6**, respectively show examples of a wide angle shaded image and a wide angle corrected image. More specifically, a wide angle gradient shaded image **610** is produced via a wide angle lens component (such as lens component **432**) which includes a normal field of view lens and a wide angle field of view lens. In certain embodiments, the shaded image is represented as a red, green, blue (RGB) image. In certain embodiments, the red, green and blue each have respective shading characteristics **612**.

[0048] In certain embodiments, a camera compensation management system (such as camera compensation system **152**) manages shading irregularities produced by the wide angle field of view portion of the lens system. In certain embodiments, managing the shading irregularities via the camera compensation management system produces a uniform shaded image **620** which includes uniform shading characteristics **622**.

[0049] In certain embodiments, the transition from a normal field of view lens portion to a wide angle field of view lens portion produces a pronounced shading irregularity at the shading boundary produced by the image generated which include both the normal field of view lens portion and the wide angle field of view lens portion. In certain embodiments, a camera compensation management system manages the pronounced shading irregularity produced by the image generated which includes both the normal field of view lens portion and the wide angle field of view lens portion.

[0050] FIGS. **7A** and **7B**, generally referred to as FIG. **7**, respectively show examples of a wide angle distorted image and a wide angle corrected image. More specifically, a wide angle distorted image **710** is produced via a wide angle lens component (such as lens component **432**) which includes a normal field of view lens and a wide angle field of view lens. In certain embodiments, a camera compensation management system (such as camera compensation system **152**) manages image distortion irregularities produced by the wide angle field of view portion of the lens system. In certain embodiments, managing the image distortion irregularities via the camera compensation management system produces a corrected image **720** which includes mitigated distortion characteristics.

[0051] In certain embodiments, the transition from a normal field of view lens portion to a wide angle field of view lens portion produces a pronounced image distortion irregularity produced by the image generated which include both the normal field of view lens portion and the wide angle field of view lens portion. In certain embodiments, a camera compensation management system manages the pronounced image distortion irregularity produced by the image generated which includes both the normal field of view lens portion and the wide angle field of view lens portion.

[0052] The present invention is well adapted to attain the advantages mentioned as well as others inherent therein. While the present invention has been depicted, described, and is defined by reference to particular embodiments of the invention, such references do not imply a limitation on the invention, and no such limitation is to be inferred. The invention is capable of considerable modification, alteration, and equivalents in form and function, as will occur to those ordinarily

skilled in the pertinent arts. The depicted and described embodiments are examples only, and are not exhaustive of the scope of the invention.

[0053] Consequently, the invention is intended to be limited only by the spirit and scope of the appended claims, giving full cognizance to equivalents in all respects.

Claims

1. An embedded camera system for a display device comprising: a camera component, the camera component being physically coupled to a display frame to embed the camera component within the display device; and, a lens component, the lens component comprising a normal field of view lens portion and a wide angle field of view lens portion.
2. The embedded camera system of claim 1, wherein: the wide angle field of view lens portion is constructed to include a metasurface component.
3. The embedded camera system of claim 2, wherein: the lens component includes a color filter glass portion; and, the metasurface component is mounted on a color filter glass portion of the lens component.
4. The embedded camera system of claim 1, wherein: an outside edge of the normal field of view lens portion is contiguous with an inside edge of the wide angle field of view lens portion.
5. The embedded camera system of claim 1, further comprising: a camera control system, the camera control system controlling a receiving field of view of the camera component.
6. The embedded camera system of claim 5, wherein: the camera control system includes a camera compensation management system, the camera compensation management system managing image irregularities produced by the wide angle field of view lens portion.
7. A display device comprising: a display component; a display frame; an embedded camera system, the embedded camera system comprising a camera component, the camera component being physically coupled to a display frame to embed the camera component within the display device; and, a lens component, the lens component comprising a normal field of view lens portion and a wide angle field of view lens portion.
8. The display device of claim 7, wherein: the wide angle field of view lens portion is constructed to include a metasurface component.
9. The display device of claim 8, wherein: the lens component includes a color filter glass portion; and, the metasurface component is mounted on a color filter glass portion of the lens component.
10. The display device of claim 7, wherein: an outside edge of the normal field of view lens portion is contiguous with an inside edge of the wide angle field of view lens portion.
11. The display device of claim 7, wherein the embedded camera system further comprises: a camera control system, the camera control system controlling a receiving field of view of the camera component.
12. The display device of claim 11, wherein: the camera control system includes a camera compensation management system, the camera compensation management system managing image irregularities produced by the wide angle field of view lens portion.
13. A system comprising: a processor; a data bus coupled to the processor; and a display device, the display device comprising a display component; a display frame; an embedded camera system, the embedded camera system comprising a camera component, the camera component being physically coupled to a display frame to embed the camera component within the display device; and, a lens component, the lens component comprising a normal field of view lens portion and a wide angle field of view lens portion.
14. The system of claim 13, wherein: the wide angle field of view lens portion is constructed to include a metasurface component.
15. The system of claim 14, wherein: the lens component includes a color filter glass portion; and, the metasurface component is mounted on a color filter glass portion of the lens component.

16. The system of claim 14, wherein: an outside edge of the normal field of view lens portion is contiguous with an inside edge of the wide angle field of view lens portion.

17. The system of claim 16, wherein the embedded camera system further comprises: a camera control system, the camera control system controlling a receiving field of view of the camera component.

18. The system of claim 17, wherein: the camera control system includes a camera compensation management system, the camera compensation management system managing image irregularities produced by the wide angle field of view lens portion.
